MILITARY SPECIFICATION

AMINE BONDING AGENT

1. SCOPE

1.1 Scope This specification covers the requirements, quality assurance provisions and packaging for the manufacture and acceptance of one type of amine bonding agent (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-286	- Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1218	- ACS Chemicals
MIL-STD-1235	- Single and Multilevel continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the of the DoDISS specified in the solicitation. The issues of documents which have not been adopted shall be those in effect on the date of the cited DoDISS.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D2905	- Statement on Number of Specimens Required to Determine the Average Quality of a Textile Material
D2196	- Rheological properties of Non-Newtonian Coatings, Test
D3951	- Standard Practice for Commercial Packaging
E168	- Recommended Practice for General Techniques in Infrared Quantitative Analysis
E300-73	- Recommended Practice for Sampling Industrial Chemicals

(Application for copies should be addressed to the American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Composition. This amine bonding agent is one type of reaction product of tetraethylenepentamine and acrylonitrile. The material shall conform to the requirements given in Table I.

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TABLE I. Chemical and Physical Properties for Amine Bonding Agent

Characteristic	Requirement Min.	Requirement Max	Applicable Test Paragraph
Secondary Amine, eq/100g	0.9	1.2	4.5.1
Total Amine, eq/100g	1.1	1.5	4.5.2
Tertiary Amine, eq/100g	0.2	0.3	4.5.3
Ammonia, ppm	---	500	4.5.4
Moisture, %	---	0.85	4.5.5
Viscosity, 25°C, cps	500	1,000	4.5.6

3.2 Infrared Spectrum. The IR spectrum of the sample shall contain the major peaks shown in Figure 1 when tested in accordance with 4.5.7.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Workmanship. The manufacturer shall implement procedures and controls to assure that the process and the product produced are not compromised by foreign materials and contaminants or any other conditions which may degrade the composition. Determination of foreign materials shall be in accordance with 4.5.8.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the

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contractor of the responsibility of assuring that all products and supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of know defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material..

4.2 Classification of inspections. The inspection requirements specified here are classified as follows:

a. First article inspection (see 4.3).

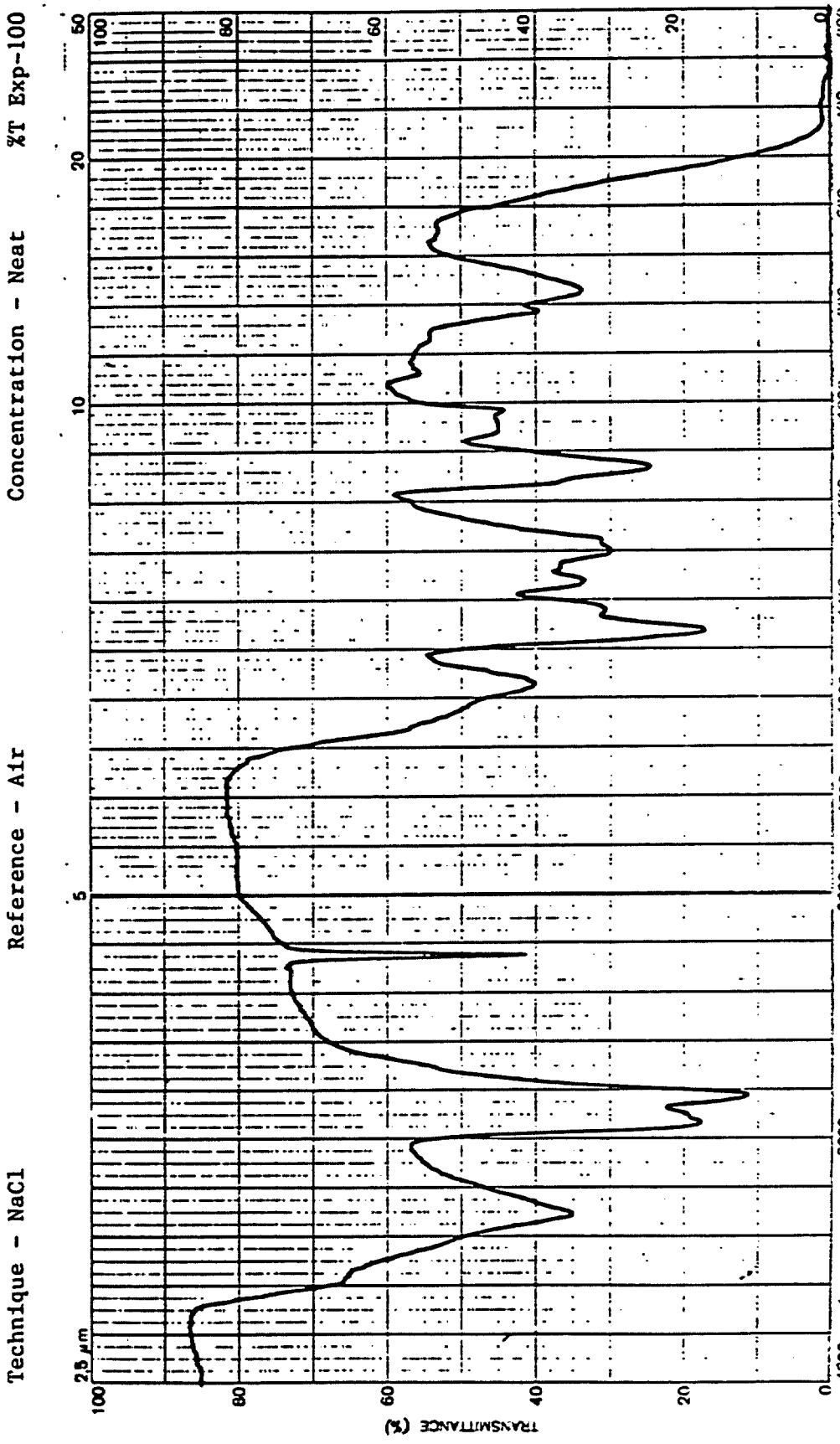
b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

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Figure 1 - IR Spectrum For Amine Bonding Agent

MAJOR PEAKS, cm^{-1}	FUNCTIONAL GROUP
3400	NH, NH_2
2950, 2820	CH_3, CH_2
2250	$C \equiv N$
1460	CH_2
1130	NH



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4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall be ten (10) pounds of amine bonding agent. The sample shall be obtained from the first production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspection to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified Table II (see MIL-A-48078).

4.3.3 Rejection. See MIL-A-48078.

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TABLE II. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-A-70604 (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Seondary amine			3.1	4.5.1
	Total amine			3.1	4.5.2
	Tertiary amine			3.1	4.5.3
	Ammonia			3.1	4.5.4
	Moisture			3.1	4.5.5
	Viscosity			3.1	4.5.6
	Infrared spectrum			3.2	4.5.7
	Workmanship			3.4	4.5.8

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-OA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Lot formation. A lot shall consist of one or more batches of amine bonding agent produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of amine bonding agent that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. The lot shall comply with the provisions for submission of product as specified in MIL-STD-105. The criteria and procedure for the assignment of lot numbers shall be in accordance with MIL-STD-1168. Also, MIL-A-48078 applies.

4.4.2 Examination. See MIL-A-48078.

a. Workmanship. Examination for workmanship shall be conducted on each sample selected for testing in accordance with 4.4.3.1. If any sample fails to meet any requirement the batch represented by the sample shall be rejected..

4.4.3 Testing. PRECAUTION: This specification covers sampling and testing of toxic and hazardous material. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing the amine bonding agent.

4.4.3.1 Sampling. Approximately 500 grams of the composition shall be selected from each batch to be sampled using ASTM Method E300-73 for liquids. Samples shall be selected for inspection in accordance with table below. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected. Each required test shall be performed in duplicate. No composite samples shall be used. The classification of defects shall be used as given in Table III.

Number of containers in batch		Number of containers to be sampled
2	100	2
100	200	5
200	300	8
300	500	10
500	1000	13
1000	3000	20
3000	10,000	32

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TABLE III. Classification of Defects

Test/Examination	Requirement Paragraph	Defect Classification
Secondary amine	3.1	Major
Total amine	3.1	Major
Tertiary amine	3.1	Major
Ammonia	3.1	Major
Moisture	3.1	Major
Viscosity	3.1	Major
Infrared spectrum	3.2	Major
Workmanship	3.4	Major

4.4.4 Inspection equipment. For the performance of all tests and examinations specified in 4.4 and 4.5, commercial inspection equipment should be employed. The contractor shall have available, and utilize correctly, this equipment and is charged with the responsibility of assuring that proper calibration procedures are followed.

4.5 Test methods and procedures. All tests given in this section shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks or ASTM D2905. Also, unless otherwise specified herein, all chemicals shall be Reagent Grade or ACS Grade in accordance with MIL-STD-1218. See 6.3 for use of equivalent test methods.

4.5.1 Secondary amine.

4.5.1.1 Apparatus.

a. Beckman pH meter or equivalent, equipped with glass-Ag/AgCl combination electrode, or automatic titrator

b. Balance, analytical, accurate to ± 0.0001 gm

c. Fume hood

d. Cylinders, graduated, 5-, 25-, and 50-ml

e. Stirrer, magnetic

f. Bottles, wide-mouth, 8-ounce or equivalent

g. Syringe, 1 ml, disposable

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4.5.1.2 Reagent.

a. Carbon disulfide, AR grade

b. Pyridine, AR grade

c. 2-Propanol, AR grade (isopropyl alcohol)

d. Standardized sodium hydroxide solution, 0.25N aqueous

e. Deionized water

4.5.1.3 Procedure.

a. Using a 1 ml syringe, weigh, by difference, 0.25 to 0.35 $\pm$ 0.0001 gm (W) of sample into an 8-ounce bottle.

b. With a graduated cylinder, add 50 ml of pyridine, 25 ml 2-propanol, and 25 ml of DI water.

c. Place on magnetic stirrer and stir to dissolve.

d. Using a 5 ml graduated cylinder, add (with stirring) 5 ml of carbon disulfide.

e. Titrate the sample with standard 0.25N aqueous sodium hydroxide using the following procedure or an automatic titrator.

(1) While stirring with a magnetic stirrer, add 1 ml sodium hydroxide to the bottle. Take the pH reading when it has stabilized. Record volume added and corresponding pH reading.

(2) Continue process. Add smaller aliquots of sodium hydroxide when nearing the equivalence point.

(3) Continue titrating until there is less than a 0.2 change in pH upon addition of 1 ml sodium hydroxide.

(4) Graph total ml sodium hydroxide added verses pH.

f. Determine the volume used to reach the potentiometric break point by determining the midpoint of the break curve. Record the volume (V_1).

g. Run two blank titrations following procedure in steps b through f. Titrate the blank at the slowest speed possible.

h. Blank volumes must agree within 0.03 ml. Record the average volume as V_2.

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i. Dispose of titrated solutions in such a way that vapors do not enter the laboratory atmosphere.

Calculation:

$$\text{Secondary amine content, eq/100 g} = \frac{(V_1 - V_2) \times N}{10 \times W}$$

Where:

V_1 = sample titration volume, ml
V_2 = Blank titration volume, ml
N = normality of sodium hydroxide
W = sample weight, gm

4.5.2 Total amine.

4.5.2.1 Apparatus.

a. Beckman pH meter, or equivalent, equipped with glass-Ag/Ag/AgCl combination electrode or an automatic titrator.

b. Balance, analytical, accurate to ± 0.0001 gm

c. Cylinder, graduated, 100 ml

d. Bottles, 4-ounce or equivalent

e. Syringe, 1 ml, disposable

4.5.2.2 Reagents.

a. p-toluenesulfonic acid solution, 0.5N. Dissolve 95 gm of p-toluenesulfonic acid monohydrate, AR grade, in approximately 900 ml of glacial acetic acid. Slowly add 20 ml of acetic anhydride with constant stirring. Dilute to one liter with glacial acetic acide and mix.

b. Glacial acetic acid, AR grade

c. Acetic anhydride, AR grade

4.5.2.3 Standardization of p-Toluenesulfonic Acid Solution.

a. Weigh, by difference, 0.9 ± 0.0001 gm (W) of primary standard grade potassium hydrogen phthalate.

b. Dissolve in a 4-ounce bottle using 50-75 ml of the glacial acetic acid.

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c. Place the sample to be titrated in a steel cup to reduce electrostatic interfence.

d. Titrate potentiometrically with the p-toluenesulfonic acid solution, using the following procedure or an automatic titrator.

(1) While stirring with a magnetic stirrer, add 1 ml p-toluenesulfonic acid solution to the bottle. Allow to stir and take the pH reading when it has stabilized. Record volume added and corresponding pH reading.

(2) Continue process, adding smaller aliquots of p-toluenesulfonic acid solution when nearing the equivalence point.

(3) Continue titrating until there is less than a 0.2 change in pH upon addition of 1 ml p-toluenesulfonic acid solution.

(4) Graph total ml p-toluenesulfonic acid solution added versus pH.

e. Determine the volume required to reach the equivalence point by locating the midpoint of the break curve. Record the volume as V.

f. Calculate the normality of the p-toluenesulfonic acid solution as follows:

$$N = \frac{W}{V \times 0.20423}$$

Where:

W = weight of potassium hydrogen phthalate
V = volume of titrant used

g. Average the duplicate results.

NOTE; p-toluenesulfonic acid solution should be restandardized every two weeks.

4.5.2.4 Procedure.

a. Using a 1 ml syringe, weigh, by difference, 0.25 - 0.30 $\pm$ 0.0001 gm (W) of sample into a 4-ounce bottle.

b. Using a graduated cylinder, add 50 ml of glacial acetic acid and stir to dissolve.

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c. Titrate the sample with the standardized 0.5N p-toluenesulfonic acid solution using the following procedure or an automatic titrator.

(1) While stirring with a magnetic stirrer, add 1 ml p-toluenesulfonic acid solution to the bottle. Allow to stir and take the pH reading when it has stabilized. Record volume added and corresponding pH reading.

(2) Continue process, adding smaller aliquots of p-toluenesulfonic acid solution when nearing the equivalence point.

(3) Continue titrating until there is less than a 0.2 change in pH upon addition of 1 ml p-toluenesulfonic acid solution.

(4) Graph total ml p-toluenesulfonic acid solution added versus pH.

d. Determine the volume required to reach the potentiometric break point by locating the midpoint of the break curve.

Calculation:

$$\text{Total amine, eq/100g} = \frac{V \times N}{10 \times W}$$

Where:

V = Volume of p-toluenesulfonic acid, ml
N = normality of p-toluenesulfonic acid
W = sample weight, gm

4.5.3 Tertiary Amine. Tertiary amine content shall be determined in accordance with the following:

a. Tertiary amine (eq/100g) = total amine eq/100g) - secondary (eq/100g).

4.5.4 Ammonia. Ammonia content shall be determined in accordance with the following:

4.5.4.1 Apparatus.

a. Ammonia gas-sensing electrode

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HNU Model ISE-10-00, HNU Systems, Inc.,
30 Ossipee Road, Newton, MA 02164

or

Orion Model 95-10, Orion Research, Inc.,
380 Putnam Avenue, Cambridge, MA 02139

or

b. Meter, pH/mV, digital, Orion Model 801A or equivalent

c. Stirrer, magnetic

d. Syringe, 3 ml, disposable

e. Cylinder, graduated, 100 ml

f. Balance, analytical, accurate to ± 0.0001 gm

g. Paper, graph, three-cycle semi-logrithmic

h. Holder, electrode, 20° angle

i. Flask, volumetric, 2000-, 1000-, 500- and 100-ml

4.5.4.2 Reagents.

a. Sodium hydroxide, 10M. Add 4 gm of AR grade NaOH to 80 ml of DI water in a 100 ml volumetric flask, dissolve, and dilute to volume with deionized water.

b. Standard ammonium chloride solution, 100-, 50-, and 10-ppm as ammonia. To prepare a 100 ppm as NH_3 standard, add 0.6288 gm of reagent grade ammonium chloride to about 1000 ml of DI water in a two liter volumetric flask. Dissolve and dilute to volume with DI water. Dilute 500 ml of 100-ppm standard to one liter with DI water to prepare a 50 ppm as HN_3 standard. Dilute 100 ml of 100 ppm standard to one liter with DI water to prepare as 10 ppm as HN_3 standard.

c. Ammonium chloride, AR grade

d. Sodium hydroxide, AR grade

e. Standard laboratory glassware

4.5.4.3 Calibration Procedure.

NOTE: Recalibrate every two hours.

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a. Assemble the ammonia electrode according to instruction manual.

b. Connect the electrode to the pH/mV meter, referring to instruction manual for specific directions.

c. Add 100 ± 1 ml of the 10 ppm as NH_3 standard to a 150 ml beaker. With stirring, add 1 ml of 10M NaOH.

NOTE: Some magnetic stirrers generate sufficient heat to change solution temperature, which affects electrode response. To minimize this effect, place a piece of insulating material such as cork, polyurethane foam, or styrofoam under the beaker.

d. Supply gentle agitation with the magnetic stirrer. After one minute, record the millivolt reading for the 10-ppm NH_3 standard. Rinse the electrode with DI water.

e. Repeat steps c and d for the 50-ppm and 100-ppm NH_3 standard.

f. Using a three cycle semi-logrithmic graph, construct a calibration curve.

4.5.4.4 Procedure.

a. Using a 3 ml syringe, weigh, by difference, 2 - 3 ± 0.001 gm (W) of the sample into a 150 ml beaker.

b. Add 100 ± 1 ml of DI water and stir to dissolve on thermally insulated magnetic stirrer.

c. Place the electrode in the sample solution.

d. Add 1 ml of 10M NaOH with stirring, and wait one minute for a stable reading. Record the display reading (R).

e. From the previously constructed calibration curve, interpolate, using R, to determine the sample solution concentration (C),

$$\text{Ammonia content, } NH_3 \text{ ppm} = \frac{C \times 100}{W}$$

Where:

C = sample solution on concentration, ppm
W = sample weight, gm

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4.5.5 Moisture. The moisture content shall be determined in accordance with Method 101.5 of MIL-STD-286.

4.5.6 Viscosity. The viscosity shall be determined in accordance with ASTM D2196.

4.5.7 Infrared Spectrum. The infrared spectrum shall be obtained using a procedure which has had prior approval of the Contracting Officer. ASTM E168 shall be used as a basis of the procedure.

4.5.8 Workmanship. Visually examine the sample for foreign matter or contamination.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.

5.1.1 Level A. - When applicable packaging shall be as specified in the contract.

5.1.2 Level B. - When applicable packaging shall be as specified in the contract.

5.1.3 Level C. - Unless otherwise specified in contract or order, packaging and packing shall be in accordance with ASTM D-3951.

5.2 MARKING. - In addition to any special markings required by the contract or order, marking shall be in accordance with MIL-STD-129 and shall include, but not limited to the following information:

LONG TERM STORAGE SHOULD BE BELOW 32°F TO PRESERVE PRODUCT QUALITY

6. NOTES

6.1 Intended use. The amine bonding agent is intended for use in propellant for the M864 base-bleed system.

6.2 Ordering data. See MIL-A-48078 (PA)

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6.3 Equivalent test methods. The test methods give in this specification are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (accuracy nd precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, AMCCOM, ATTN: AMSMC-QAR-R(D), Picatinny Arsenal, N.J. 07806-5000. This description should include, but not be limited to, the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required (see MIL-I-45208).

6.4 Subject term (key word) listing.

Propellant
Bonding agent
Projectile

Custodian
Army-AR

Preparing activity
Army-AR

(Project 6810-A016)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-70604 (AR)	AMINE BONDING AGENT

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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